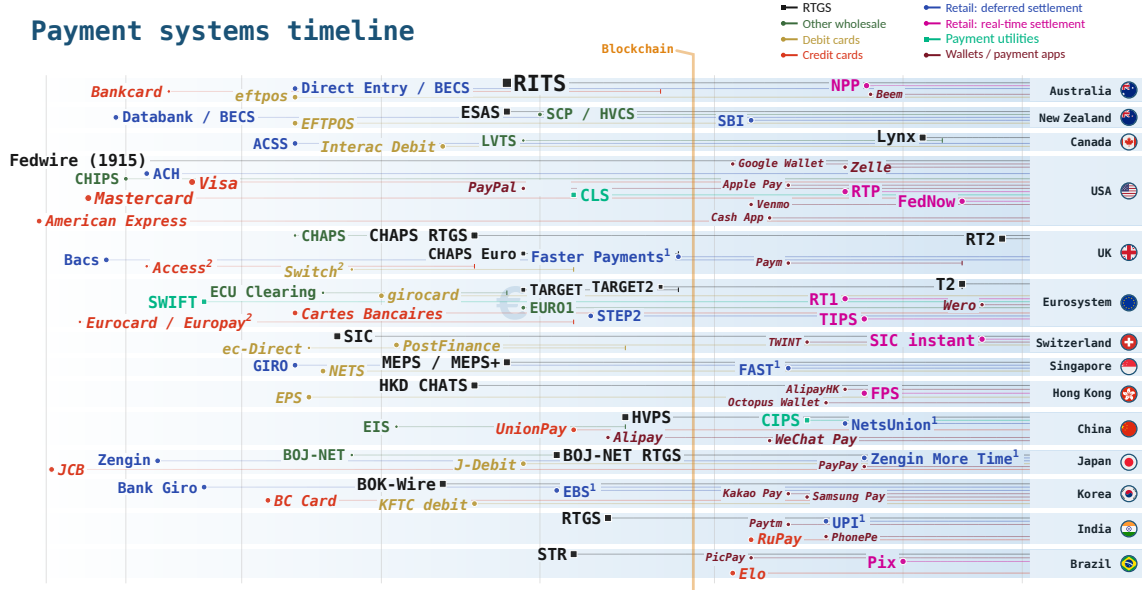
The Role of RITS
in Supporting Settlement in a Tokenised Ecosystem

Reserve Bank of Australia consultation

October 2026

Payment systems timeline



[1]Deferred interbank settlement with instant customer funds. [2]Absorbed into Mastercard: Access (1996), Eurocard/Europay (2002), Switch (2002; UK Maestro until 2011).

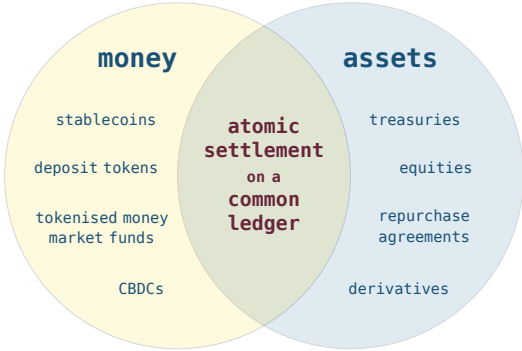
Settlement in a tokenised ecosystem

Tokenised money

Real-time gross settlement (RTGS) systems became widely adopted by central banks from the 1990s. Australia launched RTGS in 1998 with the **Reserve Bank Information and Transfer System (RITS)** as interbank settlement engine. RITS had previously settled government securities, a role that **Austraclear** took over from the RBA in 2002.

Technology limited early RTGS to wholesale payments. Retail RTGS followed in various countries over the next two decades. RITS was enhanced with the 24/7 **Fast Settlement Service (FSS)** in 2018.

Programmable ledgers support new forms of digital money, including deposit tokens and stablecoins. **Project Acacia** explored these systems and their possible links with RITS, helping inform future settlement system design.



Tokenised assets

Central securities depositories (CSDs) immobilised physical certificates and enabled book-entry transfers. Dematerialisation replaced certificates with electronic records. In Australia, **Austraclear** (1984) and **CHESS** (1994) did this for debt securities and equities.

Today, **ASX Group** operates monopoly clearing and settlement through CHESS and Austraclear. Its costly five-year effort to replace CHESS with a **Digital Asset** tokenised platform failed in 2022.

Tokenisation turns money and financial assets into programmable digital objects on a shared ledger. Contract logic governs their issuance, transfer and use. Advantages include **atomicity** of settlement operations, **composability** of automated contracts, and **interoperability** between digital infrastructures.

Tokenisation maps external legal or financial claims into programmable ledger objects. A token can be synchronised to an authoritative off-chain registry – a **digital twin** – or it can represent an authoritative registry entry itself – **native issuance**. Common token interfaces turn different kinds of financial assets into mutually addressable programmable objects.

RITS consultation areas

1. Synchronisation

How RITS/FSS could support DvP settlement of tokenised wholesale assets. Requirements for **synchronising** actions on tokenised platforms with settlement in ESA reserves.

2. At-par exchange

How RITS/FSS could support **at-par exchange** of tokenised private money issued by different institutions, and exchange between traditional bank accounts and tokenised money.

3. Reserve access

How tokenised money issuers might use **central bank reserves** if access were made available, and under what account structures and access models.

4. Tokenised reserves

Which use cases benefit from the issuance of **tokenised central bank reserves**, and the design considerations for their issuance, distribution, funding, and liquidity management.

Tokenisation

*Tokenisation in the context of the **RITS consultation** concerns settlement involving **value tokens**.*

Payment Tokens

Substitute payment identifiers

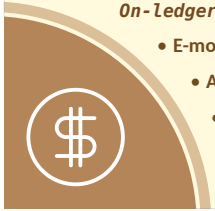
Tokenisation has different meanings within payments. **EMV payment tokens** replace a card's primary account number with a cryptographic token that can be restricted to a device or merchant, limiting misuse if stolen.



***EMV** stands for Europay, Mastercard and Visa (the three biggest card schemes before Mastercard bought Europay). **EMVCo** was established in 1999 to develop common specifications for interoperable chip-card payments.*

Value Tokens

On-ledger value or rights



- **E-money tokens:** e.g. USDC (dollar peg).
- **Asset-referenced:** asset (e.g. gold) or basket.
- **Utility:** issuer's service (e.g. storage credits).
- **Cryptocurrency:** Bitcoin, Ether.

Identity Tokens

ID claims and access rights



OpenID Connect ID Tokens identify who signed in, using JSON Web Token (JWT) format. **OAuth access tokens** authorise actions at a service. **Verifiable credentials** carry signed claims a holder presents repeatedly rather than resupplying documents. **SD-JWT**, a selectively disclosable JWT, reveals only the claims needed.

Context Tokens

Units of data for AI



AI models process text, images, audio and video as tokens. The context window limits tokens used at once. Token counts measure usage; prices reflect compute demand and supply, and may differ for input, output and cached input. Token economics will likely become central to finance as autonomous agents acquire control of value tokens.

Blockchains can record public keys to verify off-chain credentials. Identity tokens can govern who holds or transfers value tokens.

Industry ledgers

Deposit tokens and repo

Service	Instruments and ledger function	Platform	Reported activity
Kinexys Digital Payments *2019*	**Native JPMorgan deposits**: authoritative ledger; treasury payments.	Quorum	**US$1bn+/day** Payments; 2024 report
HQLA^{x} *2019*	**High-quality liquid assets**: tokenised government/corporate-bond entitlements for lending, swaps and repo; ownership changes, securities stay in custody. Canton migration planned.	Corda	**US$5bn intraday repo** First month; Aug 2025
Broadridge DLR *2021*	**Distributed Ledger Repo**: US Treasury repo, intraday repo and collateral pledges; tokenised ownership transfers, securities stay in custody. G7 collateral expansion announced Sep 2026.	Canton	**US$351bn/day** Mean repo; Aug 2026
Partior *2021*	**Interbank payments in tokenised commercial bank money** (USD/EUR/SGD); banks only, 24/7. Founders: DBS, J.P. Morgan, Temasek.	Quorum	**US$1bn+ transferred** Cumulative; Nov 2024
Fnality £FnPS *2023*	**Fnality Payment System**: native sterling wholesale balances on the authoritative ledger, backed by Bank of England funds; earmarking supports conditional payments.	Besu	**Aggregate unavailable** Payment value
Ant International *2024*	**Ant group treasury:** blockchain platform connecting partner-bank deposit services; 24/7 intra-group transfers. FX via Kinexys.	Whale	**US$600bn+ processed** Period unstated; Jan 2026
Citi Token Services *2024*	**Citi Token Services for Cash**: tokens record liabilities between Citi branches, not customer deposit claims. 24/7 liquidity transfers, subject to limits; liquidity and working-capital management.	Besu	**~US$1bn/day** Mean flow; May 2026
DBS Token Services *2024*	**DBS (Development Bank of Singapore)**: deposit tokens and programmable 24/7 treasury payments; private Ethereum Virtual Machine-compatible ledger linked to Ant's Whale.	Private EVM	**Aggregate unavailable** Payment value
HSBC TDS *2025*	**Tokenised Deposit Service**: HSBC deposit representations using private distributed ledger technology; 24/7 corporate wallet transfers. Six markets / seven currencies by Aug 2026.	Orion	**~US$28bn** Payments; Q1 2026
Standard Chartered *2025*	**Tokenised bank deposits**: SGD/USD balances on Ant's Whale platform; 24/7 intra-group treasury transfers, with additional HKD/CNH capabilities.	Whale	**Aggregate unavailable** Payment value
BNY tokenised deposits *2026*	**Bank of New York Mellon deposit mirrors**: private ledger for collateral and margin workflows; authoritative deposit records remain off-chain.	Private DLT	**Aggregate unavailable** Payment value

Industry ledgers

Securities, funds & loans

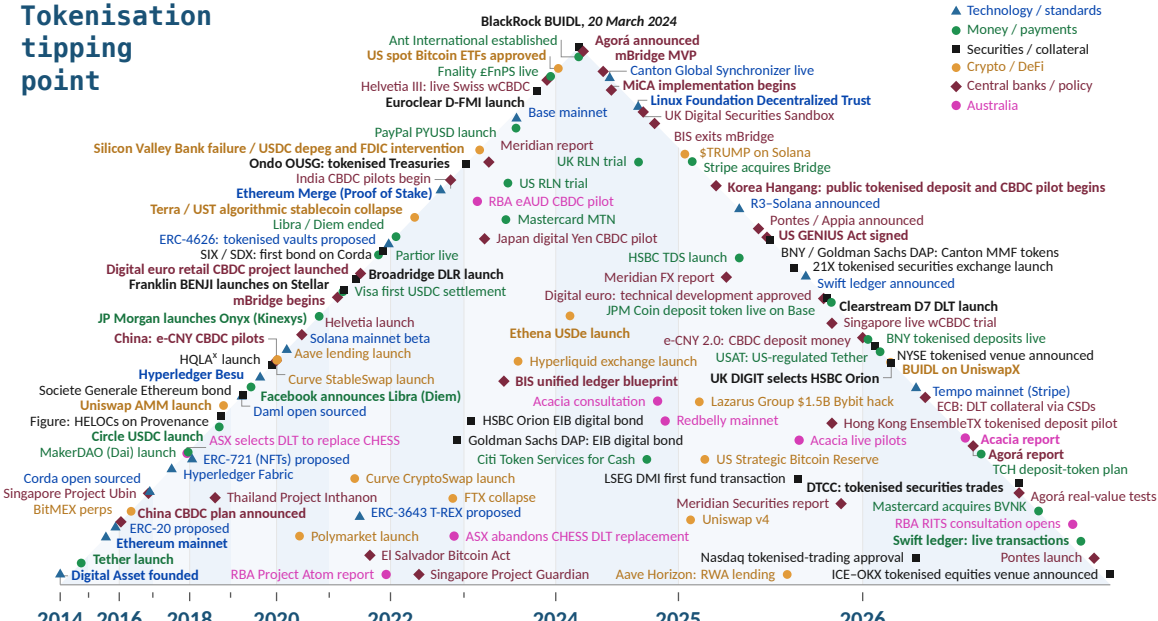
Service	Instruments and ledger function	Platform	Reported activity
Figure *2018*	**Loans, including home equity lines of credit (HELOCs)**; origination, ownership, liens and sales. DART loan register; Figure Connect trading and settlement.	Provenance	**~US$4.3bn** Loan market; Q2 2026
Ant Digital *2018*	**AntChain financial networks**: receivables, trade finance, asset-backed securities, leasing and asset tokenisation; shared records, transfers and financing.	AntChain	**32.4% market share** China BaaS; 2025
Cashlink *2019*	**Crypto-securities register**: bonds and fund units under the German Electronic Securities Act; issuance, ownership records and lifecycle management on public blockchains.	Ethereum, Polygon, Stellar	**EUR1bn+ transactions** Cumulative; Aug 2026
IZNES *2019*	**Fund-unit register**: subscriptions, redemptions, ownership and servicing; money-market, securities, real-estate and private-equity funds. Euroclear is a shareholder.	Hyperledger Fabric	**EUR32bn registered** Fund assets; May 2026
SWIAT *2020*	**Secure Worldwide Interbank Asset Transfer**: bonds, commercial paper and securities lending; appointed registrar maintains the register under the German Electronic Securities Act.	Besu	**EUR600m+ settled** Cumulative; Nov 2025
ADDX *2020*	**Private-market securities**: tokenised funds, equity and debt; issuance, custody and secondary trading on a private permissioned ledger.	Private Geth	**SGD2bn+ transactions** Total; Oct 2026
Kinexys Digital Assets *2020*	**Bonds and funds**: Digital Debt Service issuance and servicing; Fund Flow investor registers, capital calls and distributions; intraday repo and tokenised collateral transfers.	Quorum	**US$2–3bn/day** Asset transfers; Nov 2024
Clearstream FundsDLT *2020*	**Digital Transfer Agency**: blockchain fund register for issuance, investor ownership, subscriptions, redemptions and servicing; integrated into Clearstream Fund Services.	Quorum	**EUR11bn+ processed** Digital funds; Jun 2026
SIX Digital Asset Platform *2021*	**Swiss Infrastructure and Exchange**: former SIX Digital Exchange ledger for Swiss cantonal/municipal, supranational and bank bonds and private-share depositary receipts.	Corda	**CHF1.5bn+ issued** Cumulative; 2025 report
Progmat ST *2021*	**Security tokens**: real-estate trust interests, bonds and funds. Full ledger migration from Corda to Avalanche completed Jul 2026.	Avalanche	**JPY248.1bn disclosed** Issued/planned; Jul 2026

Industry ledgers

Securities, funds & loans

Service	Instruments and ledger function	Platform	Reported activity
ibet for Fin (BOOSTRY) *2021*	**Consortium securities ledger** for bonds, shares and fund interests; ownership records and transfers shared across brokers and trust banks.	ibet-Core (EVM)	**JPY133.2bn disclosed** Issued/planned; Jul 2026
Goldman Sachs DAP *2022*	**Digital Asset Platform**: native EIB / Hong Kong Government bonds; Bank of New York Mellon money-market-fund mirrors; bank keeps official books and settlement.	Canton	**Aggregate unavailable** Issuance / fund records
Euroclear D-FMI *2023*	**Digital Financial Market Infrastructure**: bank/supranational Digitally Native Notes, including World Bank; no sovereigns yet. Secondary settlement: Euroclear Bank's existing platform.	Corda	**~EUR1.2bn** Issued; FY2025 report
HSBC Orion *2023*	**Sovereign/supranational bonds**; UK Digital Gilt Instrument (DIGIT) mandate. Luxembourg: Fabric register; public Ethereum mirror tokens have no bond rights.	Hyperledger Fabric, Canton	**US$5bn+ issued** Cumulative; Jul 2026
Clearstream D7 DLT *2025*	**Commercial paper / medium-term notes**; European Investment Bank paper as Eurosystem collateral (Jun 2026). Clearstream Banking (Luxembourg): issuer central securities depository.	Besu	**Aggregate unavailable** DLT issuance
SeturionX *2025*	**Tokenised equities, bonds, exchange-traded products and tracker certificates**; formerly BX Digital. Ledger assets / Swiss Interbank Clearing cash, synchronised by the SNB's RTGS link.	Ethereum	**Aggregate unavailable** Exchange turnover
21X *2025*	**Regulated on-chain trading and settlement** of securities and fund tokens; atomic delivery-versus-payment using e-money tokens.	Polygon, Stellar	**~US$1.16m disclosed** 44 trades; Oct 2026
Axiology *2025*	**Lithuania-based debt-securities ledger**: issuance, custody, trading and settlement on a private permissioned network. Cash: euro stablecoins or central-bank money through Pontes.	XRP Ledger (private)	**~EUR17.3m registered** 7 bonds; Oct 2026
LSEG DMI *2025*	**Digital Markets Infrastructure**: private-fund tokenisation, subscriptions and distribution; connects fund managers and investors through LSEG Workspace.	Undisclosed (Azure)	**Fundraising live** Since Sep 2025
DTC Tokenization Service *2026*	**Depository Trust Company**: tokenised representations of securities held there; transfers, collateral pledges, lending and delivery-versus-payment.	Besu, Canton	**Limited production** Jul 2026; launch Oct-26

Tokenisation tipping point



AREA 1

Synchronisation

Q1. What DvP synchronisation models (asset lock, central bank reserves lock, or asset and central bank reserves lock) should RITS and/or the FSS be capable of supporting?

Q2. What types of synchronisation arrangements (asset platform-led, third-party or central bank-supported) would best support your use cases?

Q3. To what extent do existing RITS and FSS capabilities and services enable the DvP synchronisation models and synchronisation arrangements that are needed to support current and anticipated use cases?

Q4. What enhancements to the existing services and capabilities are required to support tokenised asset settlement use cases that cannot be adequately supported using existing functionality?

Central-Bank-Money-Settlement-as-a-Service

- **Delivery-versus-payment (DvP).** A trade (asset for money) can record **asset delivery** on an external platform and **payment settlement** in RITS. Each system makes its own transfers final. Removing settlement risk requires ***holding*** one, ***waiting*** for the other to settle, then ***releasing*** the hold. Payment-versus-payment (PvP) describes cross-border trades between currencies settled in different payment systems.

- **RITS already synchronises.** **Austraclear** and **CHESS** (ASX) hold securities and instruct RITS for the cash leg; **PEXA** and **Sympli** reserve funds and settle cash after confirmed lodgement; title registration remains a separate step (near-DvP). **Continuous Linked Settlement** uses its ESA to synchronise RITS with seventeen other central bank systems for PvP FX settlement. The RBA has designated **RITS and CLS** as Australia's two **Systemically Important Payment Systems** (**SIPSs**).

- **Existing RITS links are bespoke integrations.** The Bank of England has explored the alternative of a standardised **operator class** through the **Project Meridian** experiments and is developing an RTGS interface for accredited operators (2026 non-live **Synchronisation Lab**; live service targeted for 2028).

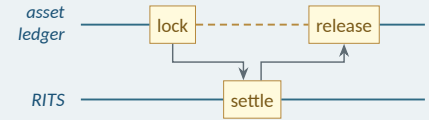
- **Limitations of today's RITS.** RITS is able to reserve funds through reservation batches until the evening cut-off; **the FSS has no reservation facility.** RITS has no common synchronisation API or credit against the asset being purchased.

- **Open question.** Whether the asset leg on a private platform has the statutory finality protection given to RITS *and also to Austraclear* under the *Payment Systems and Netting Act 1998* (PSNA).

DvP/PvP lock and release

1. Seller's asset locked (Austraclear, CHESS)

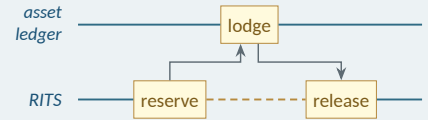
cash final; asset release pending



if the buyer can abort, the lock creates **an effective call option**

2. Buyer's reserves locked (PEXA, Sympli)

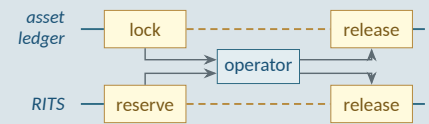
lodgement confirmed; cash release pending



if the seller can abort, the lock creates **an effective put option**

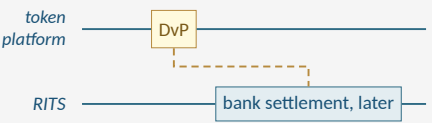
3. Dual lock (Bank of England, Pontes)

operator liveness and timeout risk



4. On-chain DvP (bank-token model)

issuer exposure until the exchange settles



Central bank synchronisation

Synchronisation Operator

Synchronisation Lab

Target: 2028

Operator earmark and release. A **synchronisation operator** instructs **RT2** to earmark the payer's funds and the asset ledger to lock the asset. When both confirm, it instructs both to complete; if either fails the holds are released. Funds move between the participants' own RT2 accounts; the operator holds no central bank money. Intended uses include securities, FX and housing. Prototyped in **Meridian (2023)** and **Meridian FX (2025)**. A Synchronisation Lab opens in 2026 for prospective operators.

Eurosystem Pontes

Launched: Sep 2026

Assets on interoperable ledgers, with launch operators **Clearstream**, **SWIAT**, **Cashlink** and **Axiology**. Connection through **Hash-Link**: the asset settles on their ledger and the cash leg is a direct T2 payment. The second route is a Eurosystem DLT cash-token platform (not yet live). **Pythagore**'s Negotiable EUropean Commercial Paper (NEU CP) market is the first planned user, from end-2026. DLT tokenised reserves mirror T2 cash (the euro RTGS).

RTGS-FMI links

Austraclear / CHESS
Securities–RITS links and batches

Fedwire Securities
Integrated securities–cash settlement

CREST / RT2
CREST cycles; earmarked sterling cash

BOJ-NET
JGB delivery-versus-payment; CSD links

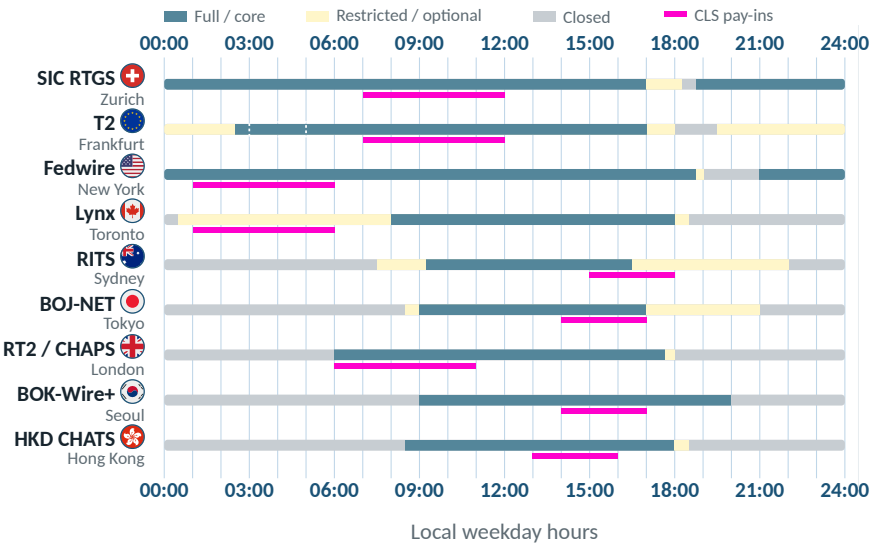
T2S
Common securities–cash platform

SECOM / SIC
Linked securities–cash settlement

CLS multicurrency PvP

Continuous Linked Settlement
75-member user-owned utility created by the international banking industry in 2002. Enables PvP settlement across 18 currencies (over US$8tn daily) within a five-hour window of overlapping RTGS hours (European morning).

RTGS wholesale operating hours



24/7 retail

SIC instant (2024)

TIPS (2018)

FedNow (2023) / RTP (2017)

Interac e-Transfer (2003) *

NPP (2018)

Zengin More Time (2018) *

Faster Payments (2008) *

KFTC EBS (2001) *

HKD FPS (2018)

* Fast customer transfers with deferred interbank settlement

Synchronisation: RITS requirements

Shorter term

- **Reservation lock on the FSS.** Continuous ***hold***, ***observe*** and ***release*** functionality on the settlement account, as in the Eurosystem's **TIPS** (retail payment system). The FSS lacks this.
- **API on RITS/FSS.** ISO 20022 already has reservation messages, `camt.046` to `camt.049`, used in the Eurosystem's **T2** and **T2S**. Operations `lock`, `status`, `settle` and `release`, with callbacks `accepted`, `expiring`, `settled` and `rejected`.
- **Defined operator class.** Authority to instruct a hold on another party's reserves and trigger their release, assuming liability for erroneous release. PEXA and Sympli do this today without infrastructure status. The Bank of England is scoping this class with UK regulators.

Longer term

- **Intraday credit inside the settlement transaction.** T2S-style **auto-collateralisation against the asset being bought** using ISO 20022 messaging. Would require compatible collateral and securities-system interfaces.
- **Admission rules.** RITS could synchronise with different types of operator, in some cases introducing financial stability risks. Formalised platform accreditation criteria must limit operator scope in line with *PFMI* standards.

PayTo does not supply a reservation, only authorisation to a payee to initiate a payment; the payer's bank then sends an ordinary NPP payment, which the FSS settles on receipt. Any hold is on the customer's account at the bank, not on ESA funds.

AREA 2

At-par exchange

Q5. What exchange models (simple exchange, exchange with synchronisation, or DLT exchange) should RITS and/or the FSS be capable of supporting?

Q6. For exchange models requiring synchronisation, what types of synchronisation arrangements (bilateral, third-party, central bank-supported or other) should be supported?

Q7. To what extent do existing RITS and FSS capabilities and services enable the exchange models that are needed to support current and anticipated use cases?

Q8. What enhancements to the existing services and capabilities are required to support exchange use cases that cannot be adequately supported using existing functionality?

Par and tokenised money

- **Monetary singleness.** Australian money is primarily held as bank deposits. Payments between deposit accounts at different banks create interbank obligations that are settled, gross or net, by **transfers between banks' reserve accounts in RITS**. This requirement limits banks' deposit creation and ensures the **fungibility of AUD deposits** at different banks.

- **Par under tokenisation.** A deposit token can circulate within one issuer's liability. Cross-issuer par exchange needs redemption and settlement arrangements, liquidity and scheme rules. Retail holders ordinarily redeem into bank deposits; they do not hold ESAs.

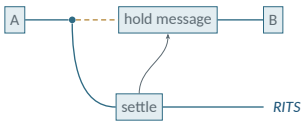
- **Project Acacia recommendation.** Bilateral convertibility becomes harder to scale as issuers and platforms multiply. A **multilateral mint-burn mechanism** linked to interbank settlement would make bank-issued deposit tokens fungible, as RITS ESA settlement does for bank deposits, supporting monetary singleness.

- **Reserves-lock design.** Requires a hold on central bank funds to support **24/7 synchronisation**, or **tokenisation of reserves**. Preferable to designs that can only lock the token or asset on the external platform.

- **Open questions.** *If RITS supports exchange between deposit tokens, should it also support exchange between other regulated money tokens (stablecoins)? Should RITS enforce par value of legally distinct instruments? Which forms of tokenised private money should be eligible for exchange through RITS/FSS? What about a regulated industry ledger, privately operated under national scheme rules?*

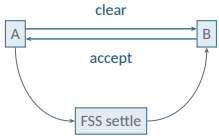
Interbank messaging and payments

Swift Y-copy (HVCS)



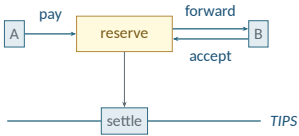
*message to the network; **settle** in RITS; then **deliver**, no acceptance step. Used for Australian domestic interbank wholesale payments (High Value Clearing System)*

Mesh (NPP)



***confirm** between institutions, then **settle**; a gateway at every participant used for bilateral clearing of payments before central bank settlement*

Hub with reservation (TIPS)



***reserve**, **forward**, **accept**, **settle**; acceptance is due seven seconds from the payer timestamp. Pix/SPI settles before forwarding; corrections use returns*

NPP Australia contracted **Swift** to build and operate a **mesh of bilateral payment gateways (PAGs)** that clear payments between banks before they are settled in the FSS. An alternative to a mesh is a **hub model**, with clearing provided by a central facility like settlement.

Hub model examples include: **TIPS** and **RT1**, **FedNow** and **RTP**, **Pix**, **Faster Payments**, **Zengin**, **SIC instant**, **SPEI**, **FPS**, the **Real-Time Rail**, **UPI**, **PayNow**, **PromptPay**, **DuitNow**, **BI-FAST** and **InstaPay**.

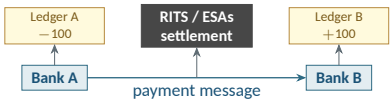
*The last six are the members of **Project Nexus**, which will connect these national retail payment systems through a **cross border hub** for real-time cross-currency retail payments.*

Australia is the only jurisdiction using a mesh infrastructure design for fast retail payments. This locks in ongoing vendor costs in the order of tens of millions annually. An enhanced FSS would be able to service the NPP under a more efficient hub design, retiring the legacy Swift PAG network.

Messaging and tokens

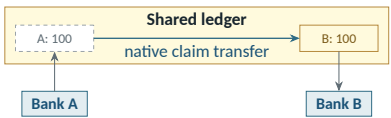
Messages

instruct a trusted intermediary to pay. Banks and the interbank settlement system (RITS) update separate ledgers.



Tokens

Native tokens transfer the recorded claim on an authoritative shared ledger. Mirror and instruction tokens can still require external settlement. Finality depends on the legally authoritative record.

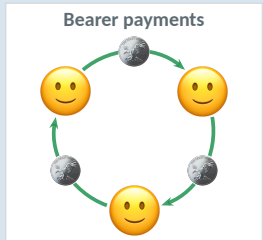


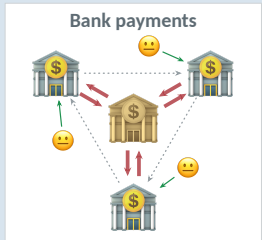
Disintermediation

A token can function as a **cryptographic bearer instrument**, a digital equivalent to physical cash (knowledge of secret keys as possession) which is held and transferred between **self-custodial** users without a bank's involvement. Like physical cash, it may be unrecoverable if lost.

A **retail CBDC** could circulate in this manner without settlement risk, likely reducing public demand for banking services. A **stablecoin** can also circulate in this manner, but is subject to credit and liquidity risk and its par value will depend on user acceptance.

Deposit token interchange preserves the position of banks as token issuers: with tokens being burned and minted in transfer (with interbank settlement support) instead of circulating freely.





ISO 20022 (2004) standardises financial messages. Swift's Cross-Border Payments and Reporting Plus (CBPR+) migration for payment instructions ran Mar-23 to Nov-25, replacing earlier MT messages. Swift is its Registration Authority and publishes the ISO 20022 data dictionary and business-process catalogue.

Money tokens

Central Bank Digital Currency

A direct claim on the central bank (i.e. government), like ESA reserves or physical cash. **wCBDC** (wholesale) settles between banks; **rCBDC** (retail) is held by households and firms. CBDCs may pay interest, as with reserves, or may not, as with banknotes and coins. As central bank money it carries no settlement risk, but as *fiat currency* central bank money it is **unsecured** with no reserve, backed only by taxation.

Stablecoin

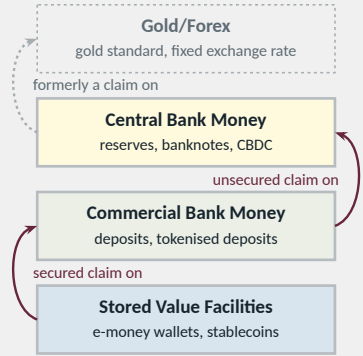
Par value claim **secured** on a pool of backing assets, stablecoins offer bankless self-custody for retail and 24/7 cross-border permissionless payments with near-instant finality. Backing assets may be a mixture of reserves (central bank money), deposits (commercial bank money) and treasury bonds or bills. Regulated stablecoins in most jurisdictions must be 100% backed by HQLA and cannot pay interest. These stablecoins are secured, however the assets securing them are not necessarily convertible to central bank money at par on demand.

An **interest-bearing** stablecoin is often a par-value fund token such as a **tokenised money-market fund**. The split is sometimes drawn as ***savings stablecoins*** (interest-paying) and ***payment stablecoins*** (no interest). A stablecoin fully-backed by central bank money is a reserves-backed digital currency (**RBDC**) or privately-issued "synthetic CBDC".

Deposit token

An **unsecured** (but sometimes government-insured), **interest-bearing** claim to central bank money issued by a commercial bank as a credit extension (*loans create deposits*). May reference a conventional deposit or be issued directly as a token. Deposit tokens are subject to the same prudential regulations as deposits (**Basel III**). Par exchange between issuers (**mint-burn interchange**) resembles payments made from conventional deposits and requires interbank settlement arrangements.

Money hierarchy



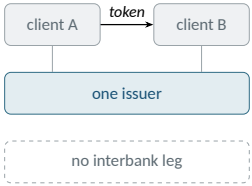
Each layer is a claim on the layer above. Lower layers carry their issuer's credit and liquidity risk; the top layer settles with finality. **RITS ensures that payments made using commercial bank money are anchored to interbank settlement with central bank money.**

Monetary singleness: an AUD$100 payment is accepted as AUD$100 irrespective of which bank issued the deposit that it was paid from. RITS settles interbank payments at par through **Exchange Settlement Accounts (ESAs)**, making Australian-dollar bank deposits fungible across banks.

Token issuance

Single issuer, one token type

token transfer

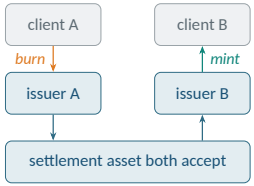


A transfer changes the holder, while the claim remains on the same issuer. "One token" means one token type, which may circulate on several chains.

Original JPM Coin system *(now Kinexys Digital Payments): private network.* ***USDT****: multiple public networks.*

Separate issuers, distinct tokens

Burn-mint interchange

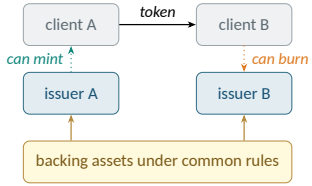


Payment extinguishes A's tokens; B's token issuance coordinated with settlement between issuers. Resembles conventional interbank payments.

Project Hangang*: banks' deposit tokens; interbank settlement in wholesale central bank money (real-value pilot).*

Common token, multiple issuers

AUD+ Scheme Proposal



Multiple legal issuers issue one fungible token under common rules. Backing, redemption and any mutual guarantee require explicit legal arrangements.

M0 / $M*: institutions admitted by protocol governance mint the same $M token against eligible collateral.*

Minting permission alone does not make an entity an issuer: the issuer carries the redemption obligation, regardless of who operates the mint. Distinct tokens are separate issuers' liabilities; exchange is at market prices unless an arrangement supports par exchange. Redemption follows the applicable issuer's terms, including eligibility and fees. A single brand can span several legal entities in one group, each with its own terms. For example, USDC is co-issued by Circle's US and French entities; redemption rights depend on the holder's jurisdiction.

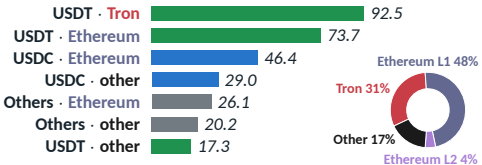
Payment Stablecoin Design Space

	Single-issuer	Multi-issuer
Off-chain reserves	• **USDT**, **Tether** (El Salvador): US$184bn; reserves held and managed by the issuer, with attestations rather than audited accounts. • **USDC**, **Circle**: US$74.9bn; segregated reserves in the Circle Reserve Fund, a registered government money-market fund. EURC EUR0.5bn. • **USDG / Global Dollar**: **Paxos** (Singapore / Europe); 150+ distribution partners share reserve earnings. Ethereum / Solana; US$3.5bn+ circulating, 3 Sep 2026. • **OUSD / Open USD**: **Bridge Building Inc.**; 200+ partners, partner board and shared reserve earnings. Base / Ethereum / Solana / Tempo; live Sep 2026. • **Qivalis**: 37-bank euro stablecoin venture; bank-owned issuer. **DNB** e-money authorisation pending; not issuing.	• **M0 / $M**: US$0.2bn Approved institutions mint against verified off-chain Treasury collateral. Other brands issue tokens backed by $M: **MetaMask mUSD**, **Noble USDN** and the **KAST Dollar**. Protocol rules set eligibility and limits; issuer obligations follow the legal arrangement.
On-chain reserves	• **USDtb**, **Ethena**: US$0.5bn; *reserves are BlackRock BUIDL tokens and stablecoins.* • **USDe**, **Ethena**: US$4.9bn. Holds ETH, BTC and stablecoins *Mixed backing / **delta-neutral hedging**: a US$1 gain on the crypto assets is offset by roughly US$1 lost on short **perpetual futures** and vice versa. Deviations from the peg may be caused by variability in funding costs, liquidations, and imperfect hedges. Direct minting and redemption are limited to approved users.*	• **GHO**, **Aave**: US$0.7bn. Created against excess collateral or in exchange for other stablecoins; Aave governance controls minting rules. • **Sky USDS / Dai**: US$11.5bn. Backed by both crypto assets and conventional financial assets.

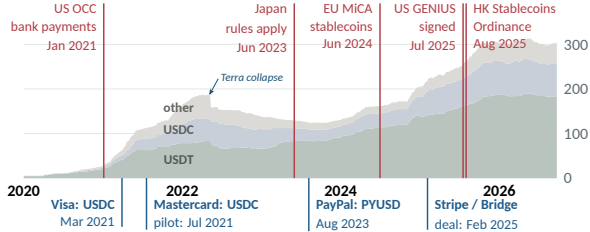
Off-chain reserves: assets in conventional custody. **On-chain reserves**: backing tokens, sometimes with off-chain underlying assets. **Single-issuer**: one issuer group. **Multi-issuer**: independent issuer groups or protocol minters.

Payment stablecoins

Coin · network circulation (US$bn)



USD-pegged supply outstanding (US$bn)



Provider	Initiative	Use cases
Visa	**USDC settlement** (Mar 2021) **Bridge cards** (Apr 2025) **Visa Direct** pilots (Sep/Nov 2025)	**Card settlement:** USDC live seven days a week; EURC pilot; PYUSD / USDG support announced; pilot expanded to nine chains (Apr 2026). **Card spending:** Bridge cards spend stablecoin balances; expanded Mar 2026. **Transfers and payouts:** Visa Direct pilots for stablecoin prefunding and wallet payouts.
Mastercard	**USDC pilot** (Jul 2021) **Settlement expansion** (Jun 2026) **BVNK** acquired (Aug 2026) **SoFiUSD** live (Sep 2026)	**Card settlement:** USDC in select markets; SoFiUSD live, with 24/7 receipts in SoFi accounts; PYUSD, USDG, USDP and RLUSD expansion announced. **Card spending:** spend stablecoin balances. **Transfers and payouts:** Mastercard Move bank withdrawals; Nuvei / Circle merchant USDC payouts announced. BVNK holds, converts and transfers bank money and stablecoins. **MTN:** programmable B2B payments and stablecoin settlement.
Stripe	**Checkout** (Oct 2024) **Financial Accounts** (May 2025)	**Checkout:** customers pay stablecoins; merchants receive fiat in their Stripe balance. **Accounts / Bridge:** receive bank or wallet transfers, hold dollar stablecoins, convert between fiat and stablecoins, and pay bank accounts or crypto wallets.
PayPal	**PYUSD** (Aug 2023) / **Xoom** (Apr 2024) **Pay with Crypto** (Jul 2025)	**PYUSD:** Paxos-issued dollar stablecoin for holding, payments and wallet transfers. **Pay with Crypto:** crypto checkout converted via PYUSD; merchants receive dollars. **Xoom:** PYUSD-to-fiat payouts over local payment rails.
Circle	**Circle Payments Network (CPN)**	**Cross-border payments:** Connects financial institutions that convert local currency into stablecoins, transfer them, and convert back for local-currency payouts.

Savings Stablecoin Design Space

	Native	Wrapped
Rebasing	• **Origin OUSD** USDC collateral earns income through lending and liquidity provision. Yield increases eligible holders' OUSD balances automatically. One token targets US$1. Token balance grows. • **BlackRock BUIDL**: Targets US$1 per token; monthly dividends arrive as additional BUIDL tokens.	• **Ondo rUSDY** Holds USDY in a wrapper contract. As USDY's reference price rises, the holder receives more rUSDY, with each rUSDY targeting US$1. The wrapper turns price growth into balance growth. • **Ondo rOUSG**: Wraps OUSG; the token balance grows. • **Aave aUSDC**: USDC lending receipt; balance grows.
Accumulating	• **Ondo USDY** A tokenised note secured by Treasury and deposit assets. Yield increases its redemption value per token; the holder's token count stays unchanged. One token represents a growing dollar value. • **Circle USYC**: Fund token; Treasury and reverse-repo income increases its token price. • **Ondo OUSG**: Fund income increases value per token.	• **Origin wOUSD** Wraps rebasing OUSD. The wOUSD balance stays fixed; each token unwraps into more OUSD as yield accrues. • **Ethena sUSDe** Deposit USDe to receive sUSDe; rewards increase the USDe recoverable per token. • **Sky sUSDS**: Deposit USDS; each sUSDS becomes redeemable for more USDS as savings accrue.

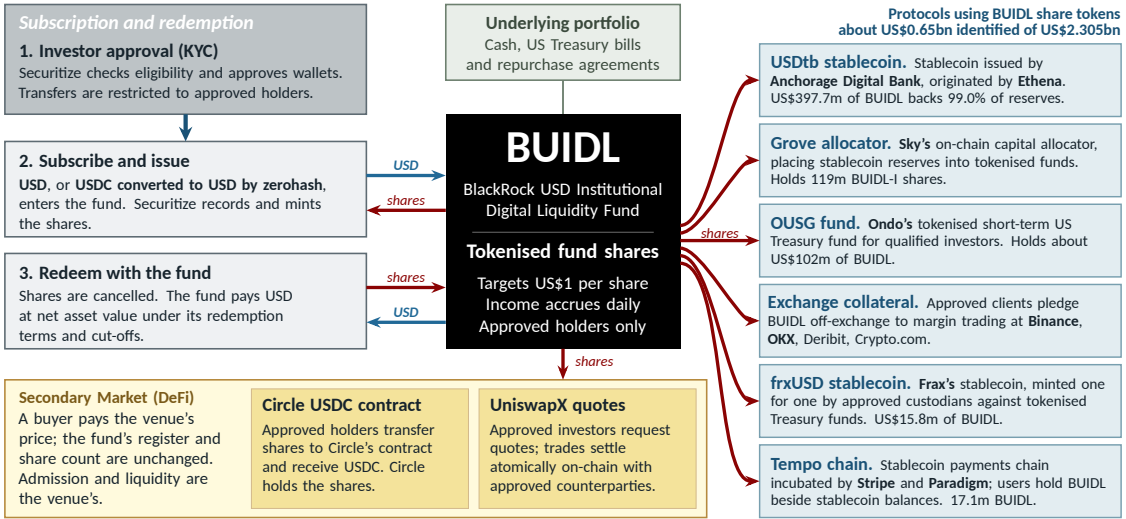
Native: yield accrues in the base token. **Wrapped**: a separate token represents underlying tokens held in a contract. **Rebasing**: yield changes the token balance. **Accumulating**: yield changes redemption value per token. The underlying legal claim can be a note, fund interest or protocol token.

Tokenised cash-management instruments

Provider / platform	Role and instruments	Selected networks	Assets US$bn	Transfers 30d	Value US$m, 30d
Ondo	Investment platform: **USDY**, secured Treasury / deposit note; **OUSG**, limited-partnership interests in money-market funds / short-term assets.	Ethereum, Stellar, Solana, XRP Ledger	**2.250** **0.396**	82,191 44	205.3 116.9
Circle	Token administrator for **USYC**: shares in Hashnote's liquidity fund, invested in Treasury bills and reverse repo; converts to USDC.	Ethereum, Solana, Canton	**2.506**	55	809.2
Benji (Franklin)	Franklin Templeton's fund-register platform: **iBENJI**, BVI institutional liquidity fund; **BENJI (FOBXX)**, US government money-market fund.	Ethereum, BNB Chain, Stellar, Solana, Aptos	**1.705** **0.672**	38 26	30.6 18.2
Securitize	Tokenisation and transfer-agent platform for **BlackRock**'s **BUIDL** USD liquidity fund; maintains investor records and supports share transfers.	Ethereum, Solana, Aptos	**2.305**	164	962.7
WisdomTree	Asset manager and fund platform: **WTGXX**, a US Treasury money-market fund, available through WisdomTree Prime and Connect.	Ethereum, Stellar, Solana	**1.230**	13	0.1
JP Morgan	J.P. Morgan Asset Management: **JLTXX**, US government money-market fund; **MONY**, private Treasury money-market fund.	Ethereum	**0.801** **0.103**	6 0	50.7 0
Libeara	Standard Chartered-backed tokenisation platform: **ChinaAMC**'s **CUMIU** digital money-market fund, Class I; **Bosera**'s **BELIF** liquidity income fund.	Ethereum	**0.554** **0.103**	0 0	0 0
Superstate	Tokenisation and transfer agent for **USTB**, **Invesco**'s short-duration Treasury fund; subscriptions, redemptions and transfers.	Ethereum, Solana, Plume	**0.571**	1,973	189.1
Centrifuge	Fund-tokenisation protocol: **JTRSY**, the **Janus Henderson / Anemoy** Treasury fund; tokenised ownership, subscriptions and redemptions.	Ethereum, Base, Stellar	**0.386**	73	1,513.4
OpenEden	Treasury-tokenisation platform: **TBILL** represents shares in a BVI liquidity fund; investors subscribe and redeem through a contract that handles purchases and redemptions of fund tokens.	Ethereum, XRP Ledger	**0.296**	23	28.6
Galaxy	Tokenisation infrastructure for **State Street**'s **SWEEP** private liquidity fund; PYUSD subscriptions and redemptions.	Solana	**0.158**	1	9.3
Spiko	Cash-management and tokenisation platform: **USTBL**, a French UCITS Treasury fund; subscriptions, ownership records and token transfers.	Ethereum, Starknet, Stellar	**0.154**	373	134.5

USDY: secured note; other instruments: fund / limited-partnership interests. BVI: British Virgin Islands; UCITS: Undertakings for Collective Investment in Transferable Securities.

BUIDL: *par, secured, yield-bearing*



BlackRock manages investments · **BNY** holds assets and administers the fund · **Securitize** maintains the share register · **zerohash** converts USDC subscriptions to USD.

Deposit token design space

	Single-bank arrangement	Multi-bank arrangement
Private ledgers	• **Kinexys Digital Payments**: J.P. Morgan *(authoritative deposit ledger)* • **HSBC TDS** *(mirrors of deposits)* • **Citi Token Services** *(Cash: interbranch liabilities)* • **DBS Token Services** *(tokenised deposit balances)* • **BNY tokenised deposits** *(mirrors of deposits)* • **Standard Chartered tokenised deposits** *(tokenised deposit balances)*	• **Partior**: participating settlement banks issue • **DeCurret DCP / DCJPY**: multi-bank design; one live issuer • **Great British Tokenised Deposit** *(Quant software coordinates payments between bank systems; live pilots 2026)* • **Hangang** *(real-value pilot)* • **Cari Network** • **Project Keystone**
Public ledgers	• **JPM Coin (JPMD), on Base** J.P. Morgan deposit liability on a public Ethereum layer-2 network; approved institutional holders. • **HSBC TDS on Canton** Simulated issuance and atomic settlement; April 2026 pilot.	• **PostFinance / Sygnum / UBS** *Swiss Deposit Token proof of concept (2025):* public chain, permissioned apps; payment instructions; cash settlement through SIC. • **CBMT public-chain route** *(testnet only)* Public testnet in the sandbox architecture, alongside private Besu chains and inter-ledger bridges.

Operator services; public chains can restrict holders. Production status and legal claims are separate. TDS: Tokenised Deposit Service; DCP: DeCurret platform; DCJPY: yen deposit token; CBMT: Commercial Bank Money Token; SIC: Swiss Interbank Clearing.

Deposit token networks

Pilots & development

Arrangement	Issuer / claim	Interbank settlement	Technology	Status / evidence
Swift ledger 17 pilot banks	Each bank issues its own deposit liabilities.	**Central-bank money** (RTGS) or **commercial-bank money** (correspondents).	Besu	Live transactions from Aug 2026; controlled pilot.
DeCurret DCP 43 pilot firms	DCJPY: yen deposit token. Issued by GMO Aozora Net Bank.	**Commercial-bank money** (lead bank) or **stablecoins** in interbank settlement tests.	DCP: DeCurret platform; Besu	Aggregate unavailable Payment value
CBMT 7 sandbox banks	Each bank issues its own test-money token.	**Central-bank money** envisaged for settling interbank balances.	Besu	Sandbox; test currency only.
Hangang 9 Phase 2 banks	Bank tokens; wholesale CBDC.	**Central-bank money:** Bank of Korea wCBDC settles between pilot banks.	Besu	Phase 1 completed; Phase 2 authorised Jul 2026.
GBTD 7 pilot institutions	Each institution issues its own deposit liabilities.	**Central-bank money:** atomic settlement explored in the BoE Synchronisation Lab.	Quant platform	Customer pilots completed Sep 2026; asset pilots next.
Cari Network 30+ joining / 6 design	Banks would issue their own deposit tokens.	**Commercial-bank settlement model:** daily net fiat settlement via a designated bank.	ZKsync Prividium (Ethereum L2)	Bank onboarding in preparation.
EnsembleTX 7 pilot banks	Banks issue tokenised deposits.	**Central-bank money:** HKD RTGS initially; tokenised central-bank money planned.	Bank ledgers	Real-value pilot through 2026.
e-CNY 2.0 2026 framework	Bank-wallet balances are commercial-bank deposits.	**Central-bank money:** PBoC settlement wallets funded with bank reserves.	Accounts + DLT	Deposit regime from Jan 2026; ongoing pilot expansion.
LSEG DiSH Bank testing	Ownership of commercial-bank deposits, recorded on DiSH.	**Commercial-bank money:** deposit ownership transfers recorded on DiSH.	Canton	Jan 2026 technical launch; repo PoCs; 2027 Partior link.
Project Keystone 5 named / 6 reported	Bank-administered claims on regulated deposits.	**Not specified:** central- or commercial-bank money for interbank settlement.	Not announced	In development; no live volume reported.
The Clearing House 17 backing banks	Banks would issue their own deposit tokens.	**Central-bank money** via planned RTP / CHIPS links; on-chain asset not yet specified.	Quant	Planned for H1 2027.

See Industry Ledgers Table 1 for mature deposit token projects. GBTD: Great British Tokenised Deposit; CBMT: Commercial Bank Money Token; DiSH: Digital Settlement House; LSEG: London Stock Exchange Group; MTN: Multi-Token Network.

Par reachability for money tokens

"Everyone can create money; the problem is to get it accepted." – Hyman Minsky

Par for money tokens means **one-for-one convertibility into central bank money**, the terminal settlement asset. RITS anchors par across Australian banks by settling interbank obligations through **Exchange Settlement Accounts** of reserve balances. The introduction of new types of par value tokens based on distinct legal arrangements complicates monetary singleness. Tokens transferred in payment must be accepted by the recipient as par-redeemable.

Tokens that transfer value through a **burn-mint mechanism** must be funded through an inter-issuer arrangement; this could settle in central bank money (equivalent to ESA balances in RITS) or commercial bank money, the latter subject to settlement risk. Fees, delays, and haircuts along the conversion path can weaken effective par and the singleness of money.



1. **Access.** The holder may lack an issuer or intermediary willing to redeem the token at par.
2. **Hours.** RITS reservations end in the evening. The FSS runs 24/7 without reservation.
3. **Capacity.** Limits on redemption throughput: intraday liquidity, batch cycles, agent bank exposure limits.
4. **Credit.** Par exchange links claims on different issuers, with different credit risks.

Ubyx "cheque" clearing model. Bank or fintech presents tokens to the issuer for redemption, then credits the customer after settlement. A token can be both a transferable issuer claim and a payment instrument. Unpaid tokens are returned; the collector does not guarantee payment. Multilateral interchangeability between different issuers' money tokens requires a multi-issuer acceptance network or **scheme**.

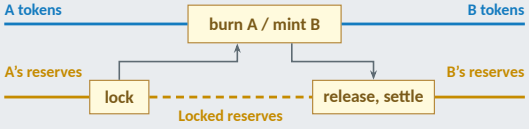
Par token interchange models

One issuer's tokens must be exchangeable for another's in the same currency at face value.

1. Sequential exchange

Burn A's tokens, transfer reserves through RITS, then mint B's tokens for the recipient.
Separate steps create timing and recovery risks.
Example: RBA simple exchange model.



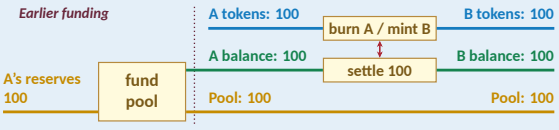


2. Reserves-lock exchange

Lock A's reserves in RITS before the token actions.
Release them to B once those actions complete.
Locks reduce settlement risk. Burn and mint must be atomic.
Example: proposed RITS reserve-lock model.

3. Prefunded utility: gross settlement

Prefund A's utility balance with pooled reserves. Each payment debits A and credits B, with customer tokens burned and minted.
Exchange can be sequential or lock A's reserve-backed balance.
24/7 settlement requires prefunded liquidity.
Settlement analogue: RTP.

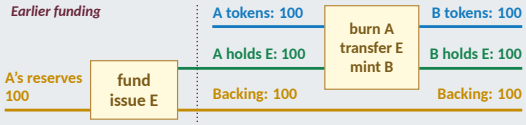


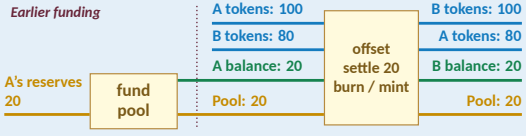
Par token interchange models

One issuer's tokens must be exchangeable for another's in the same currency at face value.

4. Backed exchange token

Prefund segregated reserves and issue E to A.
Burn A tokens, transfer E to B, then mint B tokens.
Backing stays in its reserve account, tying up liquidity.
Settlement-asset analogue: Fnality.





5. Netting: either arrangement

Queue and offset payments: A pays B 100; B pays A 80.
Both token payments complete in full; settle the net 20
using utility balances (3) or exchange tokens (4).
Netting saves liquidity; queued payments are final on release.
Settlement analogue: CHIPS.

6. Deferred interbank settlement

Exchange A tokens for B tokens and record A's debt to B.
Settle the interbank debt later through RITS, gross or net.
Payments can continue outside RTGS hours; banks carry
credit exposure until interbank settlement.
Example: Swift ledger model described by the RBA.



At-par exchange: RITS requirements

- **Membership and holder rights.** Confine par interchange to a **common instrument type** (deposit tokens). Publish risk-based issuer licensing and access criteria; specify holder rights, segregated backing, and direct ESA access or a sponsor.
- **Operator rules and redemption.** Define authority to *reserve*, *sequence*, *route* or *net* obligations. Set infrastructure supervision, finality, loss allocation and recovery rules. Standardise *hold*, *burn*, *mint*, *settlement*, *cancellation* and *status* instructions, with coordinated completion or cancellation. Provide continuous redemption, with liquidity and capacity matched to token use.
- **Settlement-token backing.** Distinguish an RBA liability, a private ESA-backed claim and a private collateralised claim. Specify the issuer, operator, insolvency treatment, collateral eligibility, haircuts, limits and loss absorption.
- **RBA policy on tokenisation and AUD settlement.** Set policy on private money, foreign-currency domestic settlement, offshore routing and open-mint liquidity. Assess the instruments made accessible and vendor concentration in custody, wallets and key management.

CPMI-IOSCO, 2022, on stablecoin arrangements: a settlement stablecoin should give holders a clear legal claim on the issuer or its reserves, be convertible at par into central bank money or other liquid assets at the latest by end of day from high-quality, segregated reserves, and have a defined, legally robust point of finality; probabilistic consensus alone is insufficient.

AREA 3

Reserve access

Q9. How might stablecoin issuers use access to central bank reserves if it was made available, both today and as arrangements scale?

Q10. What account features would be most valuable to stablecoin arrangements, both today and as they scale?

Q11. What barriers, other than eligibility, would stablecoin issuers face in accessing and using central bank reserves?

Backing, access, and remuneration

- **Reserves as backing.** Bank-deposit backing exposes the reserve holder to bank credit risk; the token holder's claim depends on the issuer's terms. Bank failure can impair recovery or delay access. Government securities carry price and liquidity risk. **Central bank money avoids commercial-bank credit risk**; redemption still needs operational access and liquidity arrangements.

- **Forte Stablecoin (Project Acacia).** An ADI held matching ESA balances under a private contract with Forte to back its **AUDF** stablecoin. Those reserves were not legally or operationally segregated from the ADI's other ESA holdings; coin holders had no claim on the RBA. ***A direct account changes the issuer's counterparty from an ADI to the RBA.***

- **Account models for consultation.** A full **ESA**; non-ADI stablecoin issuers are generally ineligible. A **settlement-only** account, as in the Fed proposal, capped and unremunerated. A **reserve-only** account transacting through an agent. An **omnibus** account on the Bank of England model, where the operator rather than the issuer holds it; CHIPS has settled through one at the Federal Reserve since 2001.

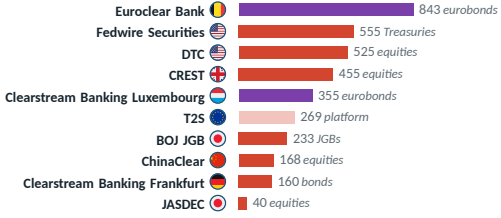
- **Remuneration.** Reserve income is a major source of issuer revenue. The Bank of England leaves stablecoin backing reserves unremunerated, treating stablecoins as payment instruments rather than savings products.

Reserve account types

The *Principles for Financial Market Infrastructures (PFMI)* permit other settlement assets where central bank money is not practical or available, subject to **strict control of credit and liquidity risk**. Controls include settlement-bank creditworthiness, exposure limits, final transfers and access to proceeds by the end of the day.

- **Exchange Settlement Account (ESA).** Normal reserves account able to support backing, payments and interchange.
- **Settlement-only account.** Payments only. Fed proposal: capped, unremunerated, no credit.
- **Omnibus account.** Pooled backing; the UK model requires a recognised, supervised payment operator.
- **No account.** Commercial bank money: industry accepts settlement risk, and a bank token (e.g. JPM Coin) can be the settlement asset.

Annual deliveries value, US$ T (2024)



Euroclear Bank and **Clearstream Banking Luxembourg** are the two *International central securities depositories* (ICSDs): depositories originally set up to settle Eurobond trades, that now settle internationally traded securities across currency areas. *Eurobonds* are debt securities issued, and largely sold, outside the domestic market of the currency in which they are denominated.

The ICSDs are among the largest financial market infrastructures in the world, holding EUR42.5 trillion and EUR21 trillion of assets under custody. Both are licensed banks that settle the cash leg in their own **commercial bank money**: the largest standing exception to the principle of settlement in central bank money.

Omnibus central bank reserve accounts

System	Legal entitlement	Settlement
Fnality £FnPS (Besu)	**Trust / beneficial ownership.** The operator holds the BoE account on trust for participants; their interests in the reserve pool are recorded on the ledger. Enabled by the BoE's April 2021 omnibus-account policy; balances are backed 1:1 by central bank funds.	DLT transfers; cash can be earmarked for delivery-versus-payment / payment-versus-payment. Live since December 2023, with statutory finality designation in December 2024.
CHIPS	**Joint account / scheme obligations.** Funding banks jointly hold the Fed account. CHIPS balances are joint obligations of those banks, payable only from the pool. CHIPS does not own the funds.	Queuing and offsetting save liquidity; payments are final on release. Fedwire funds the operating day and closing payouts.
RTP network (Vocalink)	**Joint-benefit prefunding.** Funding banks / agents share the pool; TCH controls payouts as agent. An individual RTP position is not a separate claim on the Fed.	Gross settlement, 24/7, against prefunded positions. Customers continue to hold deposits at their banks.
EURO1 (EBA Clearing) (Swift)	**Single obligation structure.** Each participant has one claim on, or obligation to, all other participants as joint creditors and obligors; all participants contribute equal shares to a liquidity pool of cash deposits held with the ECB.	Offsetting through the day; net settlement at end of day in T2. The pooled deposits are held at the ECB only during operating hours, returning to each participant until the next T2 day, where the CHIPS pool stands overnight.
RT1 (EBA Clearing) (Nexi)	**Technical account; operator as administrator.** RT1 holds a TIPS technical account at the ECB "in an administrative capacity only". The pool records Participants' positions.	Instant euro payments settle in central bank funds against prefunded positions, 24/7.
Alipay / WeChat Pay wallets	**Segregated statutory pool.** Since January 2019, 100% of customer funds sit at the PBoC, ring-fenced from the institution's creditors; wallet holders' claims are on Alipay or Tenpay, not the PBoC.	NetsUnion / UnionPay clear wallet payments; the PBoC pool funds settlement. Historical rate: 0.35% from Aug 2019; not a current-rate claim.

Each scheme has its own legal structure; full backing alone does not establish who owns the reserve balance or who owes the holder. Omnibus accounts also exist for commercial bank money settlement, for example the **Cari network**.

Tokenised repo and CoBM settlement

Tokenisation may introduce new types of reserve-holder while encouraging some existing reserve-holders to shift toward settling with *claims* on reserves (i.e. deposits). Central banks use repurchase agreements (repos) to control the supply of liquidity and influence short-term interest rates. For AUD repo conventionally settled through Austraclear, **the interbank cash payment settles in central bank money (CeBM) in RITS.**
Accepting another bank's deposit tokens settles the interbank obligation in **commercial bank money (CoBM).**
Access to and use of central bank money was explored during the RBA's Project Acacia, where one use case brought together **CBA**, Australia's largest bank; **JPMorgan**, the world's largest bank by market value, and its **Kinexys** tokenisation platform; **ASX**, operator of **Austraclear**, Australia's domestic securities depository for government debt; and **HQLAx**, a blockchain platform that transfers collateral ownership while securities remain with their custodian.

Project Acacia intraday repo



***Intraday repo**: cash lent against collateral and returned within the same day. ESA holders today obtain it from the RBA's interest-free intraday repo in RITS (unwound by end of day). Tokenised intraday repo lets banks obtain it from each other.*

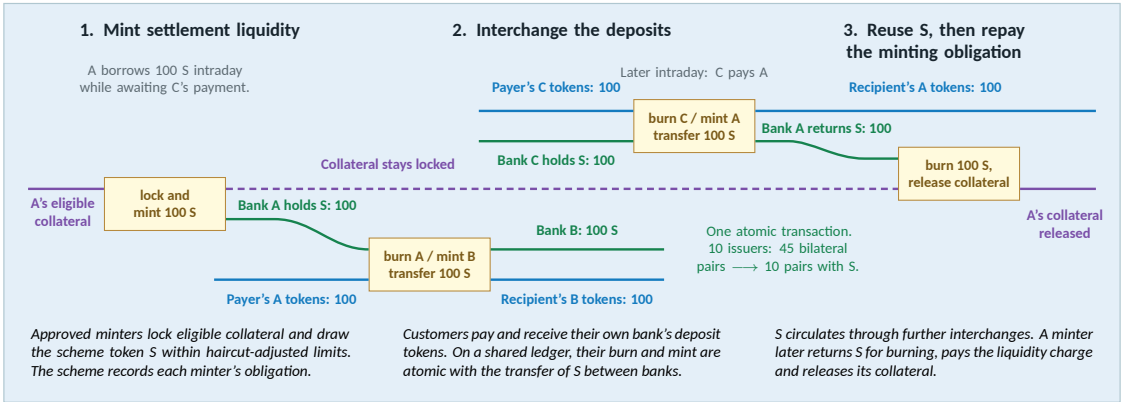
Settlement in commercial bank money. Kinexys settles with commercial bank money, not central bank money like Austraclear. CBA reportedly also demonstrated a "simulated proof of concept" where simulated "wCBDC" was used to settle an interbank obligation, but was not involved in Acacia's live pilot wCBDC experiments. The RBA has heretofore required central bank money settlement of Australian repo, per PFMI standards.

A Scheme Settlement Stablecoin

On-demand settlement liquidity against eligible (programmable) collateral

Programmability could enable a new type of reserve access: approved participants mint a common token on demand against collateral eligible for RBA domestic market operations. These privately issued balances act as "reserves" for multilateral deposit-token interchange.

An open-mint facility would apply the scheme's collateral rules (haircuts and issuance limits) automatically. Token liquidity could be priced by the amount drawn, the time outstanding, and the collateral risk. Returning the settlement token for burning retires the issuance obligation and releases the backing.



Reserve access: RITS requirements

- **Eligibility and holder protection.** Specify conditions under which licensed issuers may hold reserves, through which account and with what settlement rights.
- **Finality and limits.** State which transfers gain *PSNA* finality, whether accounts are capped, and whether non-ADI issuers will have any access to intraday credit.
- **Cash access and liquidity.** Tokens circulate continuously. Support continuous funding and redemption with issuance matched to funding; set redemption buffers, stress funding arrangements and any RBA facility access, with eligible collateral, haircuts and limits.
- **Remuneration.** Interest supports the commercial viability of full reserve backing, especially where it cannot be passed to holders. However, the central bank typically pays interest on reserves that are not costless to acquire, at rates set according to monetary policy objectives. Remuneration of reserves held against RBDC issuance would represent an effective public subsidy for narrow banking.
- **Reconciliation and reporting.** For any fully reserve-backed token, reconcile supply with segregated backing, including in-flight issuance and redemption.

UK June 2026 policy / draft rules: at least 30% unremunerated BoE reserves; up to 70% permitted short-term government assets. Liquidity backstop planned.

AREA 4

Tokenised reserves

Q12. Which use cases would become materially easier or more efficient if tokenised reserves were available?

Q13. What functional outcomes should the design of tokenised reserves, and the issuance and distribution model need to support?

Q14. Based on the use cases that require tokenised reserves and the outcomes that need to be supported, what issuance and distribution approaches would be appropriate?

Q15. What funding, redemption and liquidity-management capabilities would be required between ESAs and tokenised reserve holdings?

New forms of central bank money

- **Central bank money outside RITS.** Consultation areas 1 to 3 assume RBA central bank money stays in RITS, reaching external platforms through synchronisation or reserve access. Synchronisation references it from the platform with a hold; exchange reaches it through an interbank payment; reserve access changes the backing asset. Tokenised reserves (Area 4) place central bank money on new platforms external to RITS.

- **Tokenised central bank money (CeBM).** A **digital twin** represents a segregated ESA balance, with token supply kept aligned to that balance (Pontes design). T2 funds are escrowed in a **Token Issuance Account** under ECB legal responsibility. Matching cash tokens are claims on the ECB to transfer funds in T2; redemption returns funds to participants' T2 accounts. A native wCBDC records a direct central-bank claim on the token ledger. Issuance may still debit or segregate existing reserve balances.

- **Ledger operator.** If the RBA operates the ledger, external platforms synchronise to it as in Area 1 (this is the **Pontes** option chosen by the ECB). If a third party operates it, an RBA liability settles on infrastructure the RBA does not run (this is the **Project Helvetia** option chosen by the Swiss National Bank, which has minted wholesale CBDC on a ledger operated by SIX Digital Exchange).

- **Hours of operation.** A token can circulate continuously; conversion and liquidity between token and reserve balances will be constrained when issuance and redemption can only occur during business hours. Pontes will initially settle only within T2's operating day; continuous operation is planned for its 2028 version.

Tokenised CeBM

Design choices

Form of the token — digital twin | native mint

a twin mirrors a separate authoritative balance; a native token ledger records the central-bank claim itself

Ledger operator — RBA-operated | third-party

cash-ledger operation and oversight; same-ledger atomicity also needs assets and suitable transaction logic

Distribution — single tier | two tier

single tier: only ESA holders hold the token, for themselves; two tier: they also hold it for clients

Issuance hours — business hours | continuous

continuous circulation with limited conversion hours constrains liquidity; it need not cause a price discount

Atomicity vs control

	Liquidity pools to fund	Cash-ledger operation	Atomicity with external assets
Issued onto many platforms	one per platform, plus RITS	multiple external operators	possible, platform by platform
Issued onto one external platform	two	one external operator	possible on that platform
Issued onto RBA-operated platform	two	RBA-operated	needs cross-ledger synchronisation
No tokenised reserves issued	one	RBA-operated	needs cross-ledger synchronisation

The RBA can retain mint/burn and eligibility controls on external ledgers, as tested in Acacia. Atomic execution on one ledger requires both assets present. Cross-ledger synchronisation can also provide all-or-nothing settlement, as designed in Pontes.

21 September 2026
settles within T2's day; finality in T2

→

planned enhancement
22.5h per business day;
finality on the Eurosystem DLT

→

target: mid-2028
24/7, multi-currency and greater programmability

Pontes launched with Hash-Link and cash settlement in T2. Its planned DLT cash-token will mirror dedicated T2 reserves, leaving final cash settlement with RTGS instead of introducing a true wCBDC. For native reserves, the token ledger is the authoritative record; legal finality follows the applicable framework.

Design space for tokenised reserves

	Cash-token ledger operated by central bank(s)	Cash-token ledger operated externally
CBDC	• **mBridge** *(mBridge Ledger)*: shared central-bank platform; minimum viable product reached in 2024. • **e₹-W**: Indian wholesale pilot, since 2022. • **Hangang**: Korean central-bank-money pilot component, used as settlement asset for deposit token interchange. • **SGD Testnet**: MAS-operated shared ledger; first live SGD wholesale CBDC issuance, 2025.	• **Helvetia** *(SDX, Corda)*: Swiss National Bank sight deposits in token form on SIX; pilot extended to at least June 2028. *SIC, the Swiss franc RTGS, is run for the SNB by bank-owned SIX. Hong Kong (HKICL) and Canada (Payments Canada) also use non-central-bank operators.*
Digital Twin	• **Pontes cash-token design** *(DL3S, Hyperledger Fabric)* Tokenised central bank money (CeBM) on an ECB-operated DLT platform, representing funds in a T2 account. Cash finality remains in T2.	• **Acacia** *(Canvas Connect, Ethereum, Hedera, Redbelly, XRP Ledger)*: RBA pilot wCBDC tokens, 2026. *Although intended to model a true wCBDC, the pilot tokens were minted against conventional ESA balances that were held in reserve for the duration of their issuance.*

Synchronised RTGS settlement for interchange with no central-bank token:
initial stage EnsembleTX / HKD RTGS; UK RTGS synchronisation; SeturionX / SIC.

A multi-asset *RITS-DLT* platform?

- **Assets on the money ledger.** Europe's **Pontes DLT** (not yet live) will be able to synchronise with external platforms (via Hash Link), but will also be able to host tokenised assets *internally* and settle with *same-ledger* atomicity. This is effectively a tokenised version of **T2S**, which brought securities accounts from European depositories into the same system as cash accounts from the central banks (depositories keeping the register and legal responsibility). T2S settles securities atomically with central bank money on its own Eurosystem-owned and operated platform.
- **RITS began as a securities settlement system.** All four RITS Consultation areas assume that tokenised assets will sit externally to RITS. However, **RITS settled Commonwealth Government securities** from 1991 to 2002, before the RBA transferred that role to **Austraclear**. Other central banks serve as CSDs for sovereign debt, including the US Federal Reserve and the Bank of Japan.
- **Netting versus atomic settlement.** Where central clearing is used, trades can be netted before settlement. T2S settles each instruction gross against gross funds (**delivery-versus-payment Model 1**); CHESS settles net securities against net cash in a daily batch (**Model 3**). Austraclear uses Model 1. Atomic per-trade settlement reduces netting opportunities for these trades.
- **Gross settlement needs a liquidity mechanism.** DvP Model 1 requires liquidity on demand. T2S has **auto-collateralisation**, firing inside the settlement transaction against the securities being bought or eligible debt held. The RBA's interest-free intraday liquidity facility is member-initiated and closes with Austraclear at 6:30pm.

Project Atom, 2020–21, was an RBA test of a multi-asset ledger arrangement. A tokenised wCBDC funded, settled, and repaid a tokenised syndicated loan on an Ethereum-based platform, the CBDC and the asset on one ledger, and settlement across platforms was tested with token holds and hashed time-locked contracts.

An Orchestration Layer

A proposal to separate **core RTGS** requirements – high availability, low latency and final settlement – from a **modular overlay service**.

Liquidity Savings Mechanisms (LSMs)

LSMs optimise payment order before submission to RTGS. Most RTGS systems have queue-management systems able to offset payments multilaterally; the RITS LSM is only able to offset bilaterally.

Netting vs resequencing

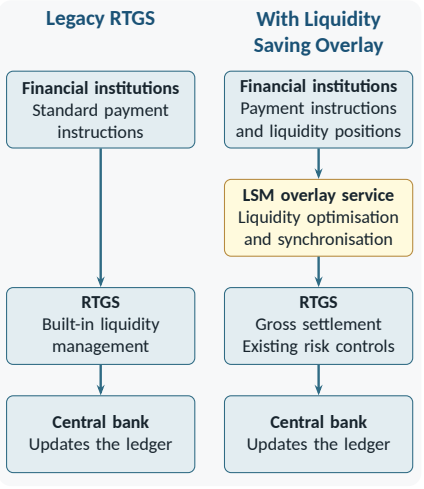
- **Netting** changes legal obligations by offsetting them, leaving only net amounts payable.
- **Resequencing** changes order, preserving every original gross obligation. All payments must still settle; the first still needs funding. Payments submitted to clear on a gross basis cannot legally be netted and batch-settled, but can be held for resequencing.

Computational difficulty

Liquidity-optimal resequencing is NP-hard in general, but polynomial-time solvable under centralised routing. Existing RTGS LSMs optimise over offset selection, not sequence selection.

Quantum optimisation of payment sequences

Hybrid quantum–classical preprocessing cut simulated liquidity needs (Bank of Canada; Bank of Italy).

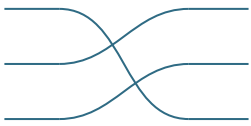


Fedwire runs pure RTGS with daylight credit and no LSM. **Gridlock resolution** covers circle search (**SIC**), bilateral offset (**RITS** auto-offset and targeted bilateral offset) and multilateral offset (**Kronos2**, **Lynx**, **CHAPS**, **BOJ-NET** and **HKD CHATS**); **T2** dissolves its queues with optimisation algorithms aimed at the highest settlement volume for the least liquidity. **CHIPS** uses stable matching, a patented (now expired) balanced-release algorithm trading liquidity savings against settlement delay.

Resequence, net, route

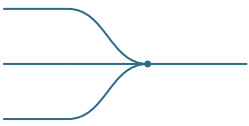
Where wholesale payments can settle or wait, **resequencing** allows substantial liquidity savings. RITS's queue offers **priority statuses** and **bilateral auto-offset** but no **multilateral offsetting algorithm** of the kind T2 (Eurosystem), CHAPS (UK), and Lynx (Canada) run. An orchestration layer that sequences an Australian dollar obligation, Swift's shared ledger, Canton's Global Synchronizer, or a central synchronisation operator for multilateral exchange, would also choose whether the obligation finally settles in RITS, or in another system — and possibly another currency.

Resequencing



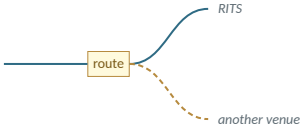
same obligations, different order, legal form unchanged

Netting



one obligation replaces three, legal form changes

Routing



same obligation, different venue; the place of finality changes

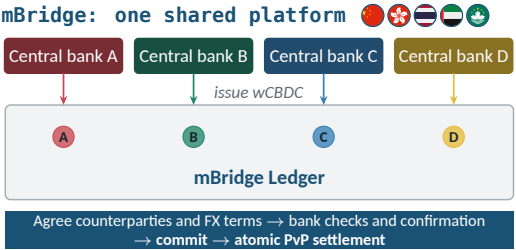
RITS settles gross and real-time, but also runs multilateral group settlement for batch and retail payment systems. Once cross-border ledgers are interoperating, domestic AUD obligations between two Australian banks could potentially be routed over an orchestration layer to settle in external multi-currency systems that the RBA does not govern. As industry shifts to a tokenised financial system that operates 24/7 across global markets, the RBA must establish whether RITS will retain its exclusive legal designation for settlement finality over domestic interbank payments.

Project Agorá

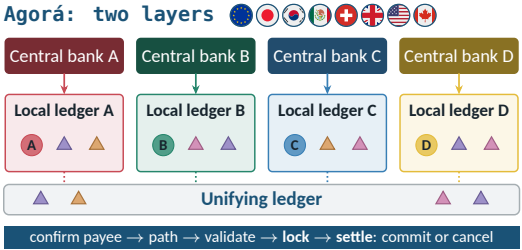
Correspondent banking routes cross-border payments through chains of **nostro accounts**, resulting in delays, limited traceability, and prefunding requirements at each intermediary. These frictions contributed to the rapid adoption of payment stablecoins, including for card scheme settlement. The **Bank for International Settlements** and various central banks have undertaken research projects to test the use of tokenised central bank money to improve cross-currency settlement.

One of these projects, **Project mBridge** (2021–present) advanced to a **Minimum Viable Product** stage in 2024.

The mBridge architecture leaves no role for correspondent banking in cross-border PvP, with settlement made in central bank money on a ledger facilitating direct settlement in CBDC. Project Agorá, launched in 2024, retains correspondent banking arrangements: each central bank's money remains on its domestic ledger, while commercial bank money circulates on a shared international ledger.



A buyer in A pays a seller in B. *The buyer's bank debits its customer's deposit and sends A's CBDC to the seller's bank, which credits the seller's deposit. The receiving bank can hold, reuse, or exchange that CBDC. If the seller wants B's currency, a bank supplies FX: the two CBDCs can be exchanged atomically through PvP.*



A buyer in A pays a seller in B. *Banks and correspondents agree the route and FX amounts, validate the payment, and lock balances. Customer and correspondent deposits change on the unifying ledger; reserves move on domestic ledgers. All legs follow one commit-or-cancel decision. The seller receives a deposit in the agreed currency.*

● Tokenised CeBM/CBDC ▲ Tokenised deposit (CoBM)

Agorá: Jurisdictional Ledger requirements

The Agorá report sets out fifteen business requirements. Each leaves scope for a variety of implementations.

1. **Jurisdictional autonomy and governance.** Central banks retain control of reserves, monetary policy, participation, and domestic operations.
2. **Issuance, redemption, and transfer.** Preserve legal claims and balance sheets, with auditable movements and exceptional issuer controls.
3. **Atomic settlement of every leg.** All legs succeed or fail together, across currencies and ledgers.
4. **Pathfinding and identification.** Agree eligible routes and verify correspondent links, respecting each bank's routing rules.
5. **Cross-currency integrity.** Apply agreed currency amounts consistently across every payment leg.
6. **Access, participation, and authorisation.** Define token access and limit delegated actions; originating participants remain accountable.
7. **Compliance pre-validation and screening.** Settle only after each bank's sanctions, AML/CFT, fraud, and balance checks pass.
8. **ISO 20022 data standards, quality, and interoperability.** Preserve payment data across systems; validate identifiers, amounts, and required fields.
9. **Reduction of exceptions and investigations.** Use early checks, clear responsibilities, timeouts, and unambiguous cancellation outcomes.
10. **Transparency, notifications, and reporting.** Give authorised participants timely status, confirmations, balances, and audit trails.
11. **Privacy and selective data-sharing.** Share only necessary data with authorised participants, within local rules.
12. **Liquidity management and monitoring.** Track reserve and deposit balances across currencies, 24/7.
13. **Governance and allocation of roles and responsibilities.** Institutions retain customer servicing, compliance, and reconciliation duties.
14. **Platform reliability, security, and scalability.** Meet measurable performance and availability standards, with predictable recovery from outages.
15. **A clearly defined settlement-finality trigger.** Define a legally enforceable finality point across jurisdictions, including in insolvency.

The Swift Ledger

Swift is the messaging network underpinning correspondent banking. Direct cross-border CBDC settlement would bypass this network, but Agorá preserves its role, positioning Swift as a natural vendor to supply its unified ledger.

Swift announced its ledger in September 2025, during Agorá's prototype development. Both use **Hyperledger Besu**. Swift calls it a "shared digital orchestration layer": coordinating payment commitments while *final* settlement uses existing systems. Oracle is integrating **Digital Assets Data Nexus** with Swift's ledger.

Tokenised reserves: RITS requirements

- **wCBDC vs "tokenised CeBM".** A digital twin can represent an ESA balance. That balance must sit in an account that cannot be spent through RITS while tokens are outstanding. Every mint or burn must be matched by a transfer into or out of it, so token supply and balance stay equal. A native token (wholesale CBDC) would be a direct RBA liability, and require its own legal basis for finality, as the RITS Regulations give ESA movements today.
- **Agorá compatibility.** A jurisdictional reserve ledger under the RBA's own authority, carrying tokenised ESA balances rather than a wCBDC, with locking, cross-domain coordination, embedded compliance, and finality under Australian law.
- **Public or private ledger operator.** An RBA-run ledger means every external platform still synchronises to it; a third-party ledger requires enforceable observation, access, governance and recovery rights, as well as clear issuer control. Any private operator would need to be domestic, credibly neutral, and user-owned; AP+ or a national industry consortium of a similar type.
- **Hours of operation.** Continuous circulation against business-hours issuance and redemption opens a conversion constraint between two RBA liabilities. The Pontes plan for continuous operation in 2028 requires tokens to remain outstanding outside T2 hours, with finality moved from T2 onto the Eurosystem's own DLT platform.
- **Intraday credit on the ledger.** Settling every payment in full requires each bank to hold its peak gross position on the ledger. RITS meets this need by intraday repo during business hours. A 24/7 ledger warrants a similar type of facility, with programmable money and assets broadening the range of possible mechanism designs.